

IoT Implementation for Air Quality Monitoring System using Low Cost Sensor

I Gusti Ayu Isnaini Fatha Ramadhani^{1,a}, Tiara Zakiyah Putri^{1,b}, Yulmaisi Dwi Asmarita^{1,c}, Sulfa Amania^{1,d}, Nandang Mufti^{2,e*}, Muhammad Alfian Mizar^{3,f}, Hasan Ismail^{3,g}, Harits Arrosyid^{4,h}

¹⁾ *Departement of Physics, Faculty of Mathematics and Natural Sciences, Universitas Negeri Malang, Malang 65145, Indonesia*

²⁾ *Centre of Advanced Materials for Renewable Energy, Universitas Negeri Malang, Malang 65145, Indonesia*

³⁾ *Departement of Mechanical Engineering, Faculty of Engineering, Universitas Negeri Malang, Malang 65145, Indonesia*

⁴⁾ *Departement of Electrical Engineering and Informatic, Faculty of Engineering, Universitas Negeri Malang, Malang 65145, Indonesia*

Email address : ^{a)}i.gusti.2303228@students.um.ac.id, ^{b)}tiara.zakiyah.2403228@students.um.ac.id,

^{c)}yulmaisi.dwi.2303228@students.um.ac.id, ^{d)}sulfa.amania.2303228@students.um.ac.id,

^{e)}nandang.mufti.fmipa@um.ac.id, ^{f)}alfianmizar@um.ac.id, ^{g)}hasan.ismail.ft@um.ac.id, ^{h)}harits.ar.ft@um.ac.id

Abstract

This research created a gas concentration monitoring system utilizing the Internet of Things (IoT), which can measure and display eight environmental parameters in real time, such as temperature, humidity, and the concentrations of gases like O₃, CO, CH₄, CO₂, SO₂, and NO₂. The system utilizes an ESP32 microcontroller to process data from sensors and transmit it to the ThingSpeak database via a Wi-Fi network. The monitoring results are displayed on a Nextion screen and an App Inventor-based application, allowing data access through smartphones. The system was tested using ambient air samples at Universitas Negeri Malang for two hours. The results indicate that the system operates stably and can accurately transmit and display data, while maintaining server connectivity even during network disruptions. The average concentrations recorded were 0.0539 ppm for CO, 323.15 ppm for CO₂, 0.35 ppm for CH₄, 0 ppm for SO₂, 0.148 ppb for O₃, and 0.0165 ppm for NO₂, indicating good air quality. The undetected SO₂ levels were due to minimal nearby sources such as traffic or industrial activity. With this success, the developed system has the potential for continuous air quality monitoring and can be integrated with other IoT platforms for further data analysis.

Keywords—Air Quality, Gas Monitoring, Industry 4.0, Internet of Things (IoT), Low-cost sensor.

I. INTRODUCTION

Air quality plays a vital role in influencing human health, the environment, economic productivity, and overall quality of life. When air quality is high in a particular area, it supports the well-being and thriving of living organisms. Conversely, poor air quality can lead to air pollution, which poses serious health risks and environmental damage [1]. The World Health Organization (WHO) has underlined the seriousness of air pollution, noting its connection to respiratory conditions, heart-related illnesses, and numerous prevalent health issues arising from breathing polluted air. [2]. In fact, air pollution ranks as the fourth leading factor contributing to declining human health [3]. According to the WHO, over seven million individuals lose their lives each year as a result of

being exposed to ambient air pollution [4]. Based on findings from the State of Global Air (SOGA), air pollution reduces the average global life expectancy by roughly 20 months [5]. As industrialization and urbanization continue to grow, air pollution levels are expected to rise due to increasing human activities and technological advancements [6].

The swift progress of the Internet of Things (IoT) has provided an effective approach to air quality monitoring. Utilizing a network of internet-enabled sensors, IoT facilitates real-time data gathering, processing, and display with improved precision [7]. A promising approach in this domain is the use of low-cost sensors integrated with Internet of Things (IoT), which enhances accessibility and affordability in air quality monitoring systems [1] [8].

The rise of Industry 4.0 has significantly boosted the use of the Internet of Things (IoT) in monitoring air quality. Defined by the fusion of advanced technologies like IoT, artificial intelligence, big data analytics, and cloud computing, Industry 4.0 has brought major changes to many industries, including the field of environmental monitoring [9]. Using IoT for air quality monitoring reflects the key principles of Industry 4.0, as it supports smart automation, continuous connectivity, and real-time data-driven decision-making [10]. By utilizing advanced data analytics and machine learning algorithms, air quality monitoring systems can detect patterns, predict pollution trends, and provide actionable insights for policymakers and urban planners [11]. Moreover, combining the Internet of Things (IoT) with Industry 4.0 technologies promotes the growth of smart cities by emphasizing innovative, data-driven approaches that prioritize environmental sustainability and the protection of public health [12].

Low-cost sensors provide a major benefit in affordability when compared to traditional air quality monitoring equipment [13]. This affordability allows for widespread implementation in urban, rural, and industrial environments [14]. Although these sensors may have limitations in accuracy compared to laboratory-grade instruments, advancements in data calibration and artificial

intelligence-based analysis have improved their reliability in providing relevant information.

In Indonesia, air quality regulations have been established through Government Regulations, the Ministry of Environment and Forestry, and the Ministry of Energy and Mineral Resources (ESDM). In 2020, the Ministry of Environment and Forestry issued Regulation Number 14 of 2020 on the Air Pollution Standard Index, replacing Ministerial Decree Number 45 of 1997 on the Calculation and Reporting of the Air Pollution Standard Index (ISPU). This updated regulation introduced revised ISPU calculations, incorporating seven key parameters: PM10 and PM2.5 particulates, nitrogen dioxide (NO₂), sulfur dioxide (SO₂), carbon monoxide (CO), ozone (O₃), and hydrocarbons (HC) [15]. When the concentration of these pollutants surpasses ambient air quality standards, they present serious health hazards, especially to the respiratory system, and may even result in life-threatening outcomes [16].

This research investigates the use of a low-cost sensor-based IoT system for monitoring air quality. The system is designed to measure various air pollution parameters, including CO₂, CO, NO₂, and PM2.5 concentrations. Data collected by the sensors will be transmitted to the cloud-based platform via wireless connectivity, enabling real-time access through web or mobile applications. Additionally, data analysis will facilitate the identification of pollution trends and provide early warnings of rising pollutant levels. By implementing this system, it is expected to raise public awareness regarding air quality and provide valuable information for authorities in formulating better environmental policies. In addition, incorporating the Internet of Things (IoT) into air quality monitoring marks an essential advancement toward building smarter and more sustainable cities. The main contributions of this study are: (1) the design and deployment of a real-time IoT-based air quality monitoring system that integrates various environmental sensors; (2) the creation of a user interface utilizing both a Nextion display and a mobile application; and (3) the evaluation of system performance through field testing under real urban environmental conditions.

II. RELATED WORK

Various studies have introduced IoT-based air quality monitoring systems employing low-cost sensors to improve accessibility and data coverage. For example, research conducted by Samad et al. developed a cloud-enabled monitoring platform that uses affordable sensors to track pollutants such as PM10, PM2.5, NO, NO₂, O₃, and CO. This system facilitates the collection of high-resolution spatial and temporal data, which is vital for effective monitoring in urban environments [17].

In addition, Hassan et al. proposed an IoT-based air quality monitoring system that incorporates blockchain technology to enhance the transparency and integrity of environmental data. The system utilizes sensors like the MQ7, DHT11 and MQ135 to measure CO, humidity, temperature, and CO₂, with the collected data securely stored on a cloud-based platform [18].

Felici-Castell et al. designed an affordable sensor network that leverages Internet of Things (IoT) technology and machine learning methods like Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM)

to forecast air quality and issue early alerts, especially aimed at protecting people with respiratory issues [19].

Furthermore, research by Johnson and Woodward introduced the Enviro-IoT system, which calibrates low-cost environmental sensors for urban environments. The study demonstrated that, with proper calibration, low-cost sensors could provide measurement accuracy comparable to standard industrial instruments [20].

Fu and Li ultimately introduced a novel AI-driven approach to decision-making for air quality monitoring, utilizing the power of big data. The model combines multiple machine learning algorithms to enhance the precision of air quality forecasts and aid in environmental protection decision-making [21].

III. EXPERIMENTAL DETAILS

This part outlines the experimental setup of the system being proposed, which include overall IoT architecture, hardware configuration, system flow, and performance testing. Each sub-section elaborates on the key components involved in designing, developing, and evaluating the IoT-based air quality monitoring system.

A. The Internet of Things (IoT) Architectural of The System

The system's Internet of Things (IoT) framework is structured into four layers, as depicted in Figure 1. The lowest layer is the perception layer, which includes the hardware components used in this study: the ESP32 microcontroller, MQ-136, MQ-131, MQ-2, MQ-135, MiCS-6814, and DHT-22 sensors. This layer functions to detect and gather environmental information. Positioned above it, the network layer manages the transfer of sensor data from the devices to the cloud server via a Wi-Fi connection. The data is then processed in the middleware layer, represented in this system by the ThingSpeak platform, which stores and manages the collected data. Finally, the application layer presents the processed information to users via a smartphone application developed using MIT App Inventor and a dedicated monitoring screen using a Nextion display.

This four-layer IoT architecture model is widely adopted in environmental monitoring systems due to its modularity and scalability [22][23]. The features of each layer are detailed in Table 1.

B. Hardware Configuration

The hardware configuration is shown in Figure 2. The +3.3V pin on the ESP32 microcontroller is connected to the positive (+) pin of the DHT-22 sensor, while the +5V pin is connected to the power pins of the MQ-136, MQ-131, MQ-2, MQ-135, and MiCS-6814 sensors. The GND pin on the ESP32 is connected to the negative (-) pin of the DHT-22 and the GND pins of the MQ-136, MQ-131, MQ-2, MQ-135, and MiCS-6814 sensors. The DHT-22 sensor's output pin is linked to pin 33 of the ESP32. Meanwhile, the analog (A0) outputs of the MQ-136, MQ-131, and MQ-2 sensors are connected to pins 32, 39, and 35 respectively, while the NO₂ output from the MiCS-6814 sensor is connected to pin 36 of the ESP32. Additionally, each analog pin of the sensors is connected to a 220 Ω resistor to ensure stable sensor readings.

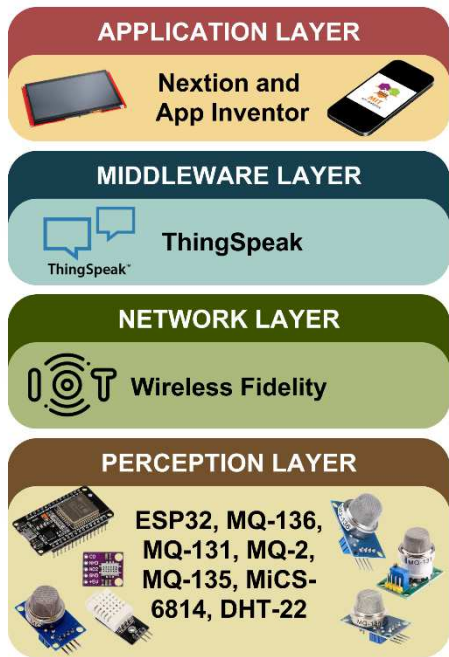


FIGURE 1 The Four-Tier Structure of the Internet of Things (IoT) Architecture

TABLE 1 The layers in Internet of Things (IoT) architecture and its feature

Layers	Devices	Features
Perception	ESP32, MQ-136 Sensor, MQ-131 Sensor, MQ-2 Sensor, MQ-135 Sensor, MiCS-6814 Sensor, and DHT-22 Sensor	Collect and Sensing Data
Network	Wifi	Forward sensor data to Middleware Layer
Middleware	ThingSpeak	Store, process, and visualization Data
Application	Nextion and App Inventor	Display sensor data real-time

C. Flowchart of The System

The system flowchart is shown in Figure 3. When the system is powered on, it begins operating by reading gas concentration levels, temperature, and humidity in the environment. The signal readings from these sensors are processed by the ESP32 microcontroller to produce measurements that comply with the standard. Once accurate data is obtained, the microcontroller attempts to connect to the internet and send the data to the ThingSpeak database for storage. If this process fails, the microcontroller will automatically reconnect to the network. The next step involves the App Inventor retrieving data from ThingSpeak to be displayed on the

smartphone application and the Nextion interface. If the data retrieval process fails, the App Inventor will continuously attempt to fetch the data from ThingSpeak until the process is successful.

D. Testing of System

The system was tested by measuring gas concentration using ambient air samples at Universitas Negeri Malang for a duration of two hours.

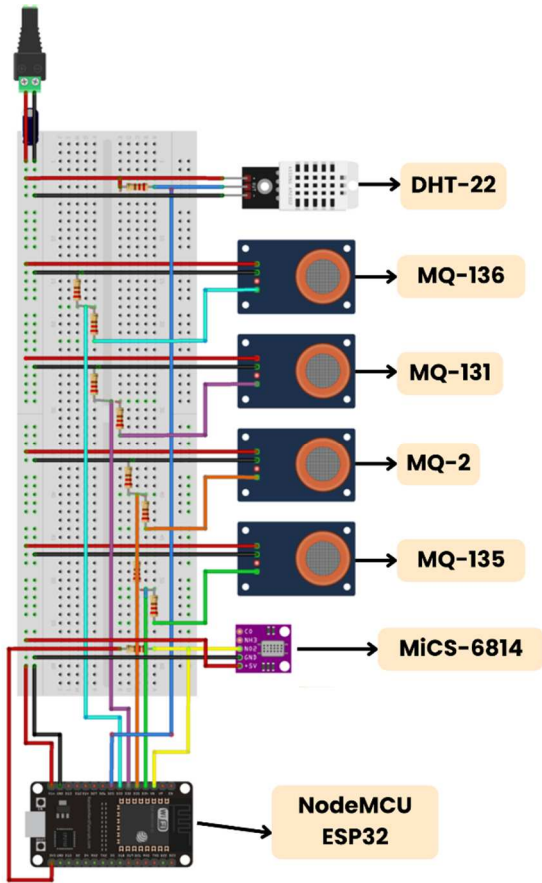


FIGURE 2 Configuration of Sensors and Microcontroller

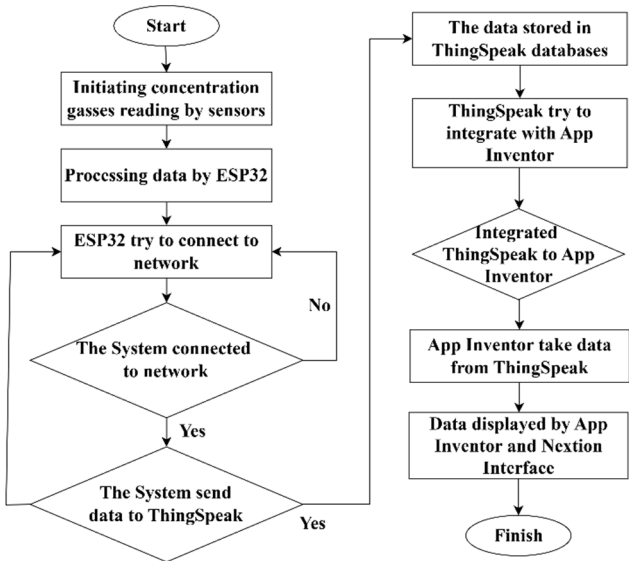


FIGURE 3 Flowchart of The Air Quality Monitoring System

IV. RESULTS AND DISCUSSION

A. Real Time Measurement Platform

In this experiment, the Nextion display and App Inventor successfully presented eight data parameters: temperature, humidity, and gas concentrations of O₃, CO, CH₄, CO₂, SO₂, and NO₂. These parameters were monitored in real-time on the Nextion display, as shown in Figure 5, and on a smartphone, as illustrated in Figure 6. This demonstrates that real-time air quality monitoring was successfully conducted using the Nextion hardware and the App Inventor software.

B. Data Validation and Analysis

According on Figure 8, it can be show that system can measure gas parameters. The average measurement of carbon monoxide (CO) gas concentration is 0.0539 ppm, carbon dioxide (CO₂) is 323.15 ppm, methane (CH₄) is 0.35 ppm, sulfur dioxide (SO₂) is 0 ppm, ozone (O₃) is 0.148 ppb, and nitrogen dioxide (NO₂) is 0.0165 ppm. All gas parameters indicate good air. Sulfur dioxide (SO₂) gas was detected at zero because motor vehicles or industry were not significant around the measurement location.

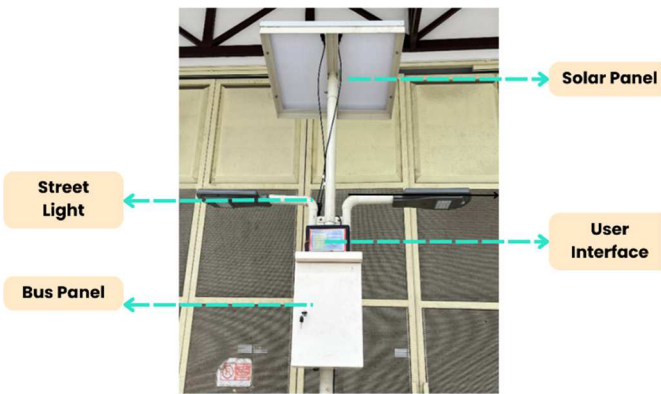


FIGURE 4 System testing using ambient air samples.



FIGURE 5 User Interface in Nextion

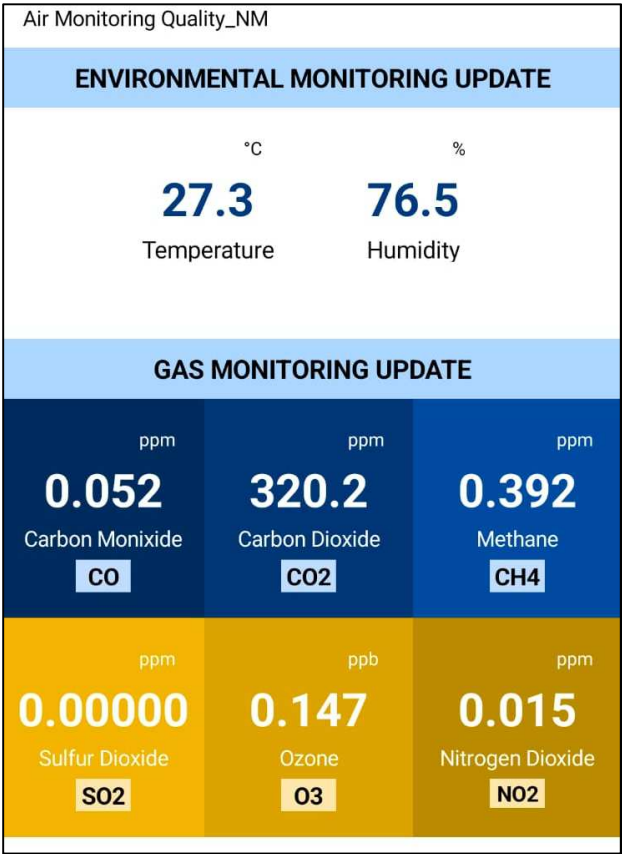
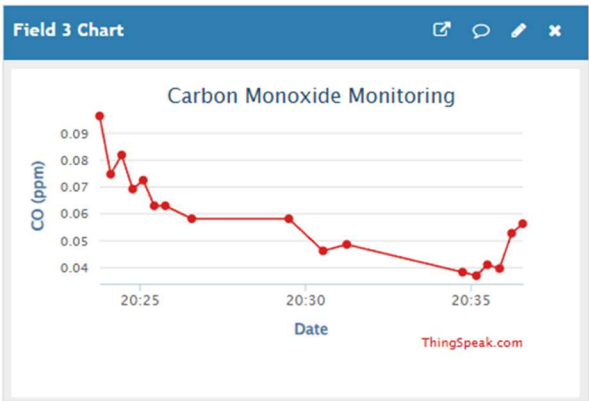
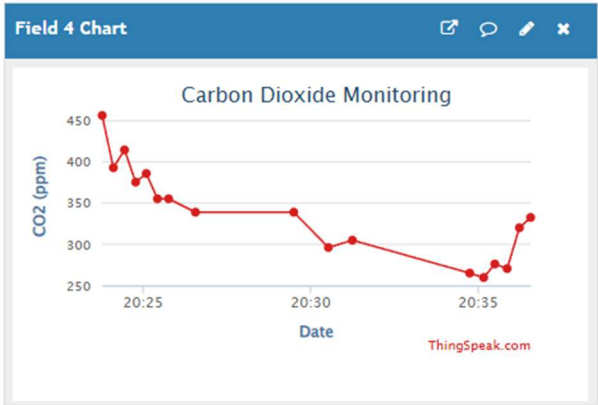


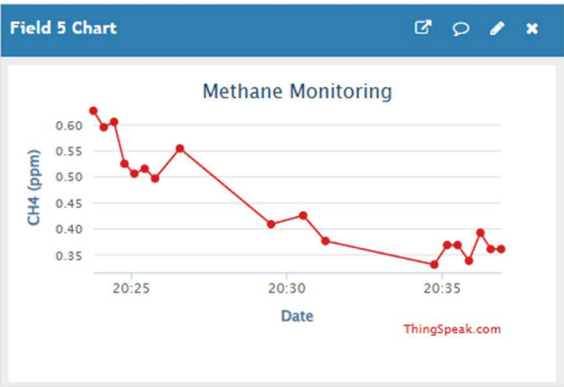
FIGURE 6 User Interface of App Inventor



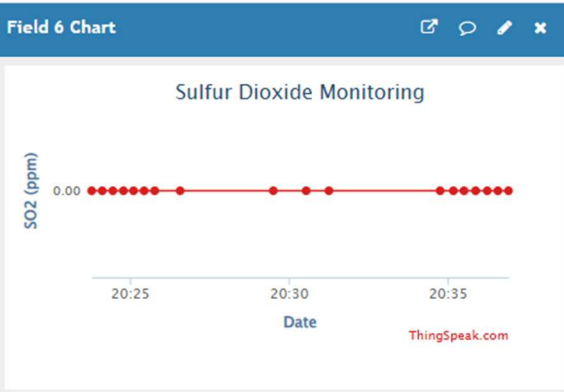
(a)



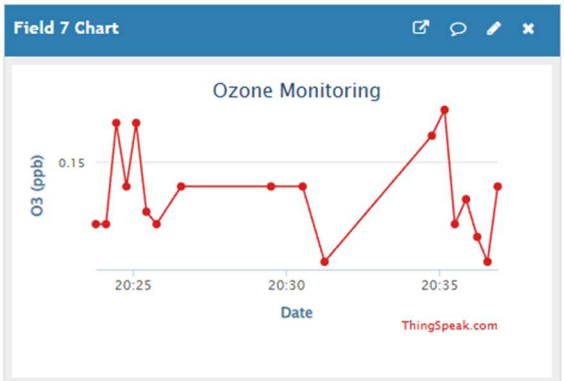
(b)



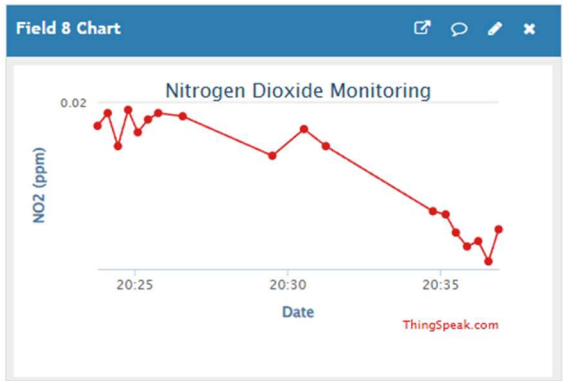
(c)



(d)

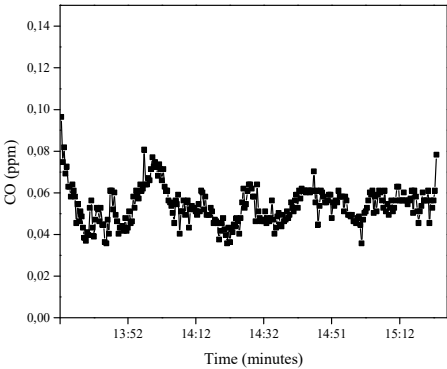


(e)

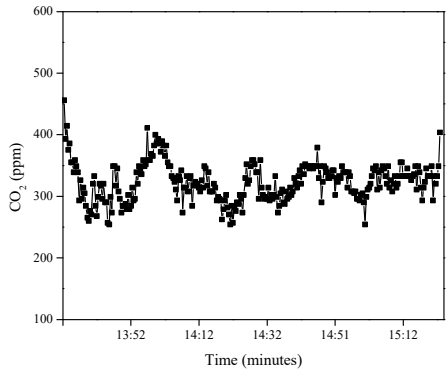


(f)

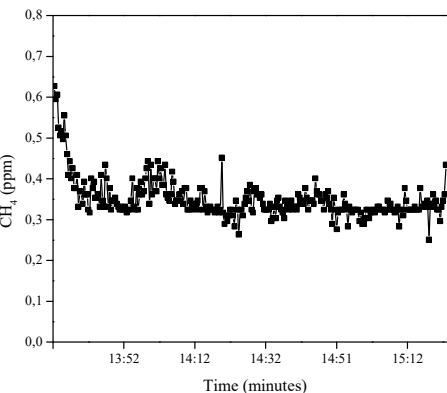
FIGURE 7 User Interface in ThingSpeak of Measuring Gas Concentrations (a) CO; (b) CO₂; (c) CH₄; (d) SO₂; (e) O₃; (f) NO₂



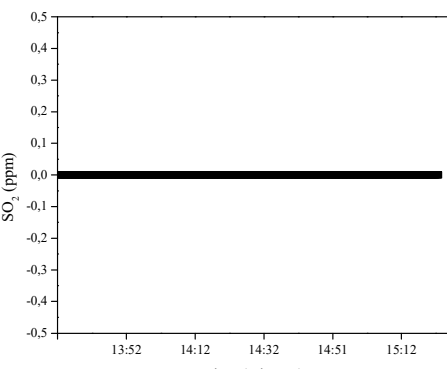
(a)



(b)



(c)



(d)

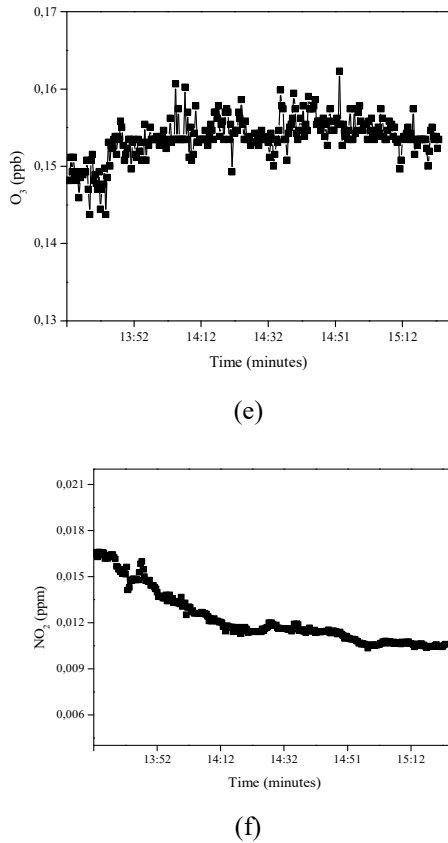


FIGURE 8. Results of Measuring Gas Concentrations (a) CO; (b) CO₂; (c) CH₄; (d) SO₂; (e) O₃; (f) NO₂ in the Air Quality Monitoring System

V. CONCLUSION

The IoT-based gas concentration monitoring system successfully measured and displayed temperature, humidity, and gas concentrations of O₃, CO, CH₄, CO₂, SO₂, and NO₂ in real-time. Data was transmitted to ThingSpeak and displayed on a Nextion screen and a smartphone application. A two-hour test showed the system operated stably, accurately, and maintained connectivity despite network disruptions. The average concentrations measured during the test were 0.0539 ppm for CO, 323.15 ppm for CO₂, 0.35 ppm for CH₄, 0 ppm for SO₂, 0.148 ppb for O₃, and 0.0165 ppm for NO₂, all of which indicate good air quality. The absence of SO₂ detection is attributed to the low presence of vehicles or industrial activities in the test area. This system has the potential for continuous air quality monitoring and further development through integration with other IoT platforms for large-scale environmental analysis. For future work, the system can be enhanced by implementing machine learning algorithms to predict air quality trends and detect anomalies, enabling more proactive and intelligent environmental monitoring.

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